Assignment: Performance Comparison of MLP, CNN, and Transformer-based Image Classifiers Using MNIST and CIFAR-10

Objective
Compare MLP, CNN and Vision Transformer classifiers on MNIST and CIFAR-10.

Tasks
1. Prepare datasets (train/validation/test, normalization, visualize samples).
2. Implement an MLP and report parameters, training time and test accuracy.
3. Implement a CNN and report the same metrics.
4. Implement a Vision Transformer (or fine-tune a pretrained ViT).
5. Evaluate using accuracy, precision, recall, F1-score, confusion matrix, parameter count, model size, training and inference time.
6. Plot learning curves, confusion matrices, correctly and incorrectly classified images.
7. Compare all models using summary tables and discuss strengths and weaknesses.

Questions
Explain why CNN outperforms MLP on images, the role of convolution/pooling, why ViT uses patches and positional embeddings, compare convergence, complexity, and recommend suitable models for handwritten digits, natural images, and resource-constrained devices.

Bonus
Study the impact of data augmentation on CNN and ViT using CIFAR-10.

Submission
Report, source code, trained models, notebook/scripts, and README.